

E-Commerce — Architecture, Marketing & Logistics

Class Notes: Classes 04–06

Part I • Section EC-2 • E-Commerce Architecture & Technologies

CO4 — Apply Internet & E-Commerce Technologies

Bloom's: Understand → Apply

Semester Final: 72 Marks

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E-Commerce Architecture & Web Technologies

CO4 Bloom's: Understand Teaching: Lecture + Diagram + Demo

Every e-commerce website — from a small Bangladeshi seller's Daraz store to Amazon's global platform — relies on a layered technical architecture. Understanding this architecture helps managers make informed decisions about technology investment, vendor selection, scalability planning, and security. This class breaks down that architecture layer by layer.

1. Multi-Tier E-Commerce Architecture

DEFINITION

A multi-tier architecture (also called n-tier or layered architecture) separates an application into distinct layers, each with a specific responsibility. Each layer only communicates with the layers directly adjacent to it. This separation makes systems easier to develop, maintain, scale, and secure.

The standard e-commerce system uses a **three-tier architecture**:

TIER 1 PRESENTATION	Presentation Layer (Client / Front-End) What the user sees and interacts with in their browser or mobile app. Built with HTML, CSS, JavaScript. Handles user interface rendering, form input, navigation, and displaying product information. Examples: the Daraz.com.bd homepage, a product detail page, a shopping cart UI. Communicates with the business logic layer via HTTP/HTTPS requests. In modern architectures, frameworks like React, Vue, or Angular build highly dynamic single-page applications (SPAs) here.
TIER 2 BUSINESS LOGIC	Business Logic Layer (Application Server / Middle Tier) The "brain" of the system. Contains all the rules that govern how data is created, displayed, stored, and changed. Processes business transactions: validates orders, calculates prices and taxes, applies discount rules, manages user sessions, handles authentication, orchestrates payment processing, triggers shipping notifications. Built with server-side languages: Python (Django/Flask), Java (Spring), PHP (Laravel), Node.js. This layer is stateless — it does not store information between requests; state is managed via sessions/tokens or passed from the client.

TIER 3

DATA LAYER

Data Layer (Database Server / Back-End)

Persistent storage of all data: product catalogs, customer accounts, order history, payment records, inventory levels, reviews. Uses relational databases (MySQL, PostgreSQL, Oracle) for structured transactional data and NoSQL databases (MongoDB, Cassandra, Redis) for unstructured, high-volume, or cached data. The database server is never directly accessible from the internet — it can only be accessed by the application server (Tier 2) for security. Data layer also includes data warehouses for analytics and reporting.

WHY SEPARATE INTO TIERS?

- ▶ **Security:** Database is isolated from direct internet access. Even if the web server is compromised, the database has an additional layer of protection.
- ▶ **Scalability:** Each tier can be scaled independently. During a sale event, you can add more application servers (Tier 2) without changing the database or front-end.
- ▶ **Maintainability:** You can update the front-end design without touching business logic. You can switch databases without rewriting application code.
- ▶ **Parallel development:** Front-end developers, back-end developers, and database administrators can work simultaneously on different tiers.
- ▶ **Reusability:** The same business logic tier can serve a web app (Tier 1 browser), a mobile app (Tier 1 phone), and a third-party API consumer simultaneously.

EXTENDED ARCHITECTURE: BEYOND THREE TIERS

Large e-commerce platforms like Amazon or Alibaba extend this to 5+ tiers:

CDN

Content Delivery Network (Edge Layer)

Static assets (images, CSS, JavaScript files) are cached at geographically distributed edge servers worldwide. A user in Dhaka requesting a product image does not wait for it to travel from a server in Virginia — it comes from the nearest CDN node, perhaps in Singapore. Reduces latency dramatically. Akamai, Cloudflare, AWS CloudFront are major CDN providers.

LOAD BALANCER

Load Balancing Layer

Distributes incoming requests across multiple web/application servers to prevent any single server from being overwhelmed. Uses algorithms: Round Robin (distribute evenly), Least Connections (send to least-busy server), IP Hash (same user always goes to same server). Essential during high-traffic events (Daraz 11.11 sale). AWS Elastic Load Balancer, NGINX, HAProxy are examples.

WEB SERVER

Web Server Layer

Handles HTTP/HTTPS request-response for static content; passes dynamic requests to application servers. Apache HTTP Server and NGINX are the dominant web servers globally. NGINX also commonly serves as a reverse proxy and load balancer simultaneously.

APP SERVER Application Server Layer (Business Logic)

Executes business logic as described in Tier 2 above. Typically runs many instances in parallel. Containers (Docker) and orchestration (Kubernetes) allow rapid scaling up/down based on demand.

DATABASE Database Layer (Persistence)

As described in Tier 3 above. At scale, uses read replicas (multiple copies of the database for read operations), sharding (splitting data across multiple databases by key), and caching layers (Redis/Memcached) in front of the primary database to reduce load.

2. Roles of Web Server, Application Server, and Database Server

Component	Primary Role	What It Handles	Common Technologies
Web Server	Serves HTTP/HTTPS requests; delivers static content	HTML pages, images, CSS files, JavaScript files, file downloads. Receives requests from browsers; sends responses. For dynamic content, passes requests to the application server.	Apache HTTP Server, NGINX, Microsoft IIS
Application Server	Executes business logic; generates dynamic content	User authentication/authorization, order processing, price calculation, inventory checks, session management, API request handling. Generates dynamic HTML or JSON responses.	Tomcat (Java), Node.js, Gunicorn (Python), PHP-FPM
Database Server	Persistent storage and retrieval of structured data	Stores all application data: users, products, orders, payments, reviews. Handles queries (SELECT, INSERT, UPDATE, DELETE). Ensures data integrity through ACID properties (Atomicity, Consistency, Isolation, Durability).	MySQL, PostgreSQL, Oracle DB, MongoDB, Redis

3. HTTP/HTTPS Request–Response Cycle

HTTP — HYPERTEXT TRANSFER PROTOCOL

HTTP is the foundation of data communication on the World Wide Web. It is an application-layer protocol that defines how messages are formatted and transmitted between web browsers (clients) and

web servers. HTTP is stateless — each request is independent; the server does not retain any memory of previous requests.

STEP-BY-STEP HTTP REQUEST-RESPONSE CYCLE

1 DNS Resolution

User types "www.daraz.com.bd" in browser. Browser queries DNS (Domain Name System) servers to translate the domain name into an IP address (e.g., 203.0.113.45). DNS is the internet's phonebook. This lookup is cached locally for subsequent requests.

2 TCP Connection (Three-Way Handshake)

Browser establishes a TCP connection with the server: (1) Client sends SYN packet, (2) Server responds SYN-ACK, (3) Client confirms with ACK. This establishes a reliable, ordered communication channel. For HTTPS, a TLS handshake follows (see below).

3 HTTP Request Sent

Browser sends an HTTP request to the server. The request includes: (a) **Method** — GET (retrieve data), POST (submit data), PUT (update), DELETE (remove); (b) **URL/Path** — which resource is being requested; (c) **Headers** — metadata: Content-Type, Accept-Language, User-Agent, Authorization token, Cookie; (d) **Body** — data payload (for POST/PUT requests, e.g., form data, JSON). Example: GET /products/laptop-hp-123 HTTP/1.1
Host: daraz.com.bd

4 Server Processing

Web server receives request → routes to appropriate application server → application server executes business logic → queries database if needed → generates response (HTML page or JSON data). For a product page: fetch product details, check inventory, retrieve reviews, compute price with current discounts, render HTML template.

5 HTTP Response Sent

Server sends response back: (a) **Status code** — 200 OK (success), 301 Moved Permanently (redirect), 404 Not Found, 500 Internal Server Error; (b) **Headers** — Content-Type, Content-Length, Set-Cookie, Cache-Control; (c) **Body** — the HTML, JSON, image, or file being returned.

6 Browser Rendering

Browser receives HTML, parses it, discovers additional resources (CSS, JS, images), sends additional HTTP requests for each, builds the DOM (Document Object Model), applies CSS styling, executes JavaScript, and renders the final visual page. A typical e-commerce product page triggers 50–200 HTTP sub-requests.

STATELESSNESS AND SESSIONS

HTTP is **stateless** by design — every request is independent; the server forgets the previous request. This is a problem for e-commerce (shopping cart must persist across pages). Solutions:

- ▶ **Cookies:** Small text files stored in the browser, sent with every request to that domain. The server sets a cookie with Set-Cookie header; the browser returns it with every subsequent request. Used for session IDs, user preferences, tracking. bKash login status is maintained via a session cookie.
- ▶ **Sessions:** Server-side storage of user state (cart contents, login status) identified by a session ID stored in a cookie. More secure than storing all state client-side because sensitive data stays on the server.
- ▶ **JWT (JSON Web Tokens):** Modern approach — server issues a signed token containing user identity and permissions. Client sends this token in every request header. Server validates the token's signature without looking up a database. Stateless but authenticated. Used in modern SPA and mobile app architectures.
- ▶ **localStorage / sessionStorage:** Browser-side storage for JavaScript apps to persist state without cookies.

HTTPS — HYPERTEXT TRANSFER PROTOCOL SECURE

HTTPS is HTTP with encryption provided by TLS (Transport Layer Security, formerly SSL). All data exchanged between browser and server is encrypted, so it cannot be read if intercepted (man-in-the-middle attacks). Absolutely mandatory for any page that handles passwords or payment information. Visual indicator: padlock icon and "https://" in browser address bar. Google penalizes non-HTTPS sites in search rankings. PCI-DSS compliance (required for accepting card payments) mandates TLS 1.2+.

4. Web Services: REST vs. SOAP

WEB SERVICES

A web service is a software system that allows different applications to communicate with each other over a network using standardized formats. It exposes business functionality as a callable interface over the internet. Web services are the infrastructure that enables e-commerce integrations: payment gateways, shipping APIs, product catalog feeds, and social login all use web services.

Feature	REST (Representational State Transfer)	SOAP (Simple Object Access Protocol)
Type	Architectural style / set of principles	Protocol with strict specification
Data format	Primarily JSON; also XML, plain text	XML only (mandatory, verbose)
Transport	HTTP/HTTPS exclusively	HTTP, SMTP, TCP — protocol-independent
Complexity	Simple, lightweight — easy to implement and consume	Complex — requires WSDL (Web Services Description Language) files, strict message structure

Feature	REST (Representational State Transfer)	SOAP (Simple Object Access Protocol)
Performance	Faster — JSON is more compact than XML; less overhead	Slower — XML parsing is heavier; envelope adds bloat
Security	HTTPS + OAuth2/JWT — flexible security layering	WS-Security built-in — enterprise-grade, standardized
State	Stateless by principle	Can be stateful or stateless
Error handling	HTTP status codes (200, 400, 404, 500)	SOAP fault messages within the XML envelope
Use cases	Public APIs, mobile apps, modern web services, microservices	Enterprise integrations, banking, healthcare, legacy systems that require contractual interfaces
Examples	Twitter API, Google Maps API, bKash payment API, Stripe API, Daraz Open Platform	Bank SWIFT messaging, HL7 healthcare, legacy SAP integrations

JSON VS. XML DATA FORMATS

Aspect	JSON (JavaScript Object Notation)	XML (eXtensible Markup Language)
Readability	Human-readable, compact, concise	More verbose — opening and closing tags for everything
Size	Smaller payload → faster transmission	Larger payload due to tag repetition
Native support	Native JavaScript object; parsed trivially in browsers	Requires XML parser in most languages
Schema validation	JSON Schema (optional)	DTD / XSD — rigorous, enforceable contracts
Metadata / attributes	No attribute concept — all data in key-value pairs	Supports element attributes, namespaces, processing instructions
Use in e-commerce	Most REST APIs, product catalog feeds, mobile apps	Legacy EDI systems, SOAP web services, some shipping APIs

5. API-Driven Commerce

An **API** is a set of rules and protocols that allows different software applications to communicate with each other. In e-commerce, APIs enable integration between the merchant's platform and third-party services without either party needing to understand the internal code of the other. APIs are the connective tissue of modern e-commerce.

KEY E-COMMERCE API CATEGORIES

- ▶ **Payment APIs:** Integrate external payment processors without handling raw card data. Merchant's website sends order details to the payment API; the API handles card authorization, fraud detection, and returns a success/failure response. Examples: *Stripe API* (global), *bKash Payment Gateway API* (Bangladesh — merchants can accept bKash payments on their own websites), *SSL Commerz* (Bangladesh's most widely used payment gateway, integrating cards, bKash, Nagad, internet banking in a single API). Security: Payment APIs use tokenization — the actual card number is never stored on the merchant's server.
- ▶ **Shipping / Logistics APIs:** Provide real-time shipping rates, generate shipping labels, create pickup requests, and track shipments programmatically. Examples: FedEx API, DHL API, *Paperfly API* (Bangladesh), *Pathao Courier API*. A merchant's checkout page can query multiple carrier APIs simultaneously to show the cheapest/fastest option.
- ▶ **Product Catalog APIs:** Allow merchants to list products on multiple marketplaces simultaneously. Daraz Open Platform API lets merchants push product listings from their own inventory system to Daraz. A single product update propagates to all connected channels automatically.
- ▶ **Inventory APIs:** Synchronize stock levels across multiple sales channels (website, Daraz, physical POS). When a product sells in-store, the API decrements inventory across all connected online platforms to prevent overselling.
- ▶ **Social Login APIs:** OAuth2-based APIs from Google, Facebook, or Apple allow users to log in with their existing social accounts instead of creating a new password. Reduces friction; increases conversion rates. Example: "Continue with Google" button on Chaldal.
- ▶ **Tax Calculation APIs:** Automatically calculate correct VAT/tax rates based on buyer location, product category, and applicable regulations. Avalara, TaxJar. In Bangladesh, VAT rules vary by product category and NBR regulations.
- ▶ **Review / Rating APIs:** Third-party review platforms (Trustpilot, Google Reviews) provide APIs to display verified reviews on merchant websites.

BANGLADESH CONTEXT — SSL COMMERZ

SSL Commerz is Bangladesh's leading payment gateway, used by 25,000+ e-commerce merchants. It aggregates all local payment methods — Visa/MasterCard, bKash, Nagad, Rocket, internet banking, and mobile banking — into a single REST API. A merchant integrates one API and automatically supports all

Bangladeshi payment methods. This "aggregator" model dramatically reduces integration complexity for local entrepreneurs.

6. Content Delivery Network (CDN)

CDN DEFINITION

A **Content Delivery Network (CDN)** is a geographically distributed network of proxy servers and data centers that delivers static web content (images, videos, CSS, JavaScript) to users from the server physically closest to them, minimizing latency and maximizing bandwidth.

HOW CDN WORKS — STEP BY STEP

- ▶ **Origin server:** The merchant's primary web server where the master copy of all content lives (e.g., in Dhaka or AWS Singapore region).
- ▶ **CDN edge nodes (Points of Presence — PoPs):** CDN providers operate hundreds of servers worldwide. Akamai has 350,000+ servers in 130 countries. Cloudflare has 300+ PoPs.
- ▶ **First request:** When a user in London first requests a product image, the CDN edge node in London does not have it — it fetches from the origin server and caches a copy at the London PoP. This is called a "cache miss."
- ▶ **Subsequent requests:** The next user in London requesting the same image gets it directly from the London CDN node — a "cache hit." Latency drops from 200ms (cross-continental) to 10ms (local PoP).
- ▶ **Cache invalidation:** When the merchant updates a product image, they must invalidate the old cached version on CDN nodes so users see the updated image. CDN providers offer APIs for programmatic cache purging.

BENEFITS OF CDN FOR E-COMMERCE

- ▶ **Performance:** Faster page loads → higher conversion rates. Amazon found that every 100ms of added page load time costs 1% in revenue.
- ▶ **Availability:** If the origin server goes down, cached CDN content is still served. CDN provides partial resilience against outages.
- ▶ **DDoS protection:** CDNs absorb Distributed Denial of Service attacks by distributing traffic across their massive global network. Cloudflare is also a major DDoS mitigation provider.
- ▶ **Bandwidth cost reduction:** Serving content from CDN edge nodes reduces the bandwidth load (and therefore cost) on the origin server.

- **Global reach:** A Bangladeshi e-commerce company can serve international customers with low latency using CDN, without operating data centers abroad.

7. Scalability: Horizontal vs. Vertical Scaling and Load Balancing

SCALABILITY

Scalability is the ability of a system to handle increasing amounts of work by adding resources. For e-commerce systems, this means the website should continue performing well whether there are 100 users or 1,000,000 users simultaneously — especially critical during flash sales, shopping festivals, or marketing campaigns.

Aspect	Vertical Scaling (Scale Up)	Horizontal Scaling (Scale Out)
Approach	Add more power to a single machine: more CPU cores, more RAM, faster SSD storage	Add more machines to the pool: distribute load across many servers working in parallel
Analogy	Upgrading one employee's tools to make them work faster	Hiring more employees to divide the work
Cost	Expensive at high end — high-end servers cost exponentially more; diminishing returns	More cost-effective — commodity servers are cheap; can add/remove as needed
Limit	Hard physical limit — a single machine can only be so powerful	Theoretically unlimited — keep adding servers
Downtime	Requires downtime to upgrade hardware	Can add/remove servers with zero downtime (rolling deployment)
Complexity	Simple — same code runs on a bigger machine	Complex — requires distributed system design, load balancing, shared state management
Failure risk	Single point of failure — if the one server fails, everything is down	High availability — one server failure doesn't bring down the system
Use case	Databases (harder to distribute), early-stage startups, legacy systems	Web/application servers, microservices, cloud-native e-commerce, Daraz's infrastructure

LOAD BALANCING

Load balancing is the technique of distributing incoming network requests across multiple servers to ensure no single server is overwhelmed. It is the operational mechanism that makes horizontal scaling work.

- **Round Robin:** Requests are distributed to servers in sequential order — Server 1, Server 2, Server 3, Server 1, Server 2... Assumes all servers have equal capacity. Simple and predictable.

- ▶ **Least Connections:** Sends each new request to the server currently handling the fewest active connections. Better for requests with variable processing times.
- ▶ **IP Hash:** The client's IP address is hashed and mapped to a specific server. The same user always reaches the same server. Important for applications where session state is stored on the server (not in a shared cache).
- ▶ **Weighted Round Robin:** Servers are assigned weights proportional to their capacity. A server with twice the RAM gets twice as many requests.
- ▶ **Health Checks:** The load balancer continuously checks if servers are alive and responsive. Failed servers are automatically removed from the pool; recovered servers are added back.

REAL-WORLD EXAM SCENARIO

Daraz's 11.11 (Singles Day) sale sees a 10x spike in traffic compared to an average day. To handle this: (1) CDN serves all static assets (product images, CSS) from edge nodes — reduces origin server load by 70-80%; (2) Auto-scaling adds new application server instances (horizontal scaling) automatically as traffic rises; (3) Load balancer distributes requests across the expanded server pool; (4) Database read replicas handle the surge in product browsing queries. This combination is how modern e-commerce handles traffic spikes without crashing.

EXAM TIP

A common exam question: "Explain the three-tier architecture of an e-commerce system with a diagram." Know all three tiers by name (Presentation, Business Logic, Data), what each contains, and why they are separated. Also be able to distinguish REST vs. SOAP, JSON vs. XML, and horizontal vs. vertical scaling with concrete examples.

Digital marketing is how e-commerce businesses acquire customers and retain them profitably. Unlike traditional broadcast advertising, digital marketing is measurable, targetable, and optimizable in real time. This class covers the major channels, the quantitative metrics that define success, and the analytical frameworks used to optimize customer acquisition and retention.

1. Digital Marketing Channels

1A. SEARCH ENGINE OPTIMIZATION (SEO)

SEO DEFINITION

SEO is the practice of optimizing a website to rank higher in organic (unpaid) search engine results pages (SERPs) for relevant queries. High organic ranking drives "free" traffic — the merchant does not pay per click. However, SEO requires significant ongoing investment in content creation and technical optimization. Results accumulate over months or years.

On-Page SEO factors (elements within the website under the merchant's direct control):

- ▶ **Keyword research and placement:** Identifying the exact words and phrases users type when searching for your products ("buy laptop Bangladesh," "best price iPhone 15 Dhaka") and placing them strategically in page titles, headings (H1, H2), meta descriptions, product descriptions, and URL slugs. Keyword stuffing (excessive repetition) is penalized.
- ▶ **Title tags and meta descriptions:** The title tag (shown as the blue link in Google results) and meta description (the two-line preview) determine click-through rate. Compelling, keyword-rich titles dramatically increase organic traffic.
- ▶ **Content quality:** Google's algorithm rewards comprehensive, accurate, well-structured content that satisfies user intent. "Thin" pages with little content rank poorly. Long-form product guides and blog posts attract organic links from other sites.
- ▶ **URL structure:** Clean, descriptive URLs (/laptops/hp-pavilion-15) rank better and are more trustworthy than parameter-heavy URLs (/product?id=4532&cat=2).
- ▶ **Page speed:** Google uses Core Web Vitals (LCP, FID, CLS) as ranking signals. Faster pages rank higher. CDN, image compression, and code minification improve speed.
- ▶ **Mobile optimization:** Google uses mobile-first indexing — your mobile site's quality determines your ranking, not the desktop version. Mobile-responsive design is mandatory.

- ▶ **Structured data / Schema markup:** Machine-readable annotations (JSON-LD format) that tell Google what your content represents — Product, Review, FAQ, BreadcrumbList. Enables rich snippets (star ratings, price, availability) in search results, boosting CTR.

Off-Page SEO factors (signals from outside the website):

- ▶ **Backlinks (inbound links):** When other reputable websites link to your pages, Google interprets this as a vote of authority and trust. A link from The Daily Star's website to a Bangladeshi e-commerce site is worth far more than a link from a low-traffic blog. The number and quality of backlinks is the strongest off-page ranking signal.
- ▶ **Domain Authority (DA):** A metric (developed by Moz) measuring the overall trustworthiness of a domain based on its backlink profile. Older, well-linked domains rank more easily for competitive keywords.
- ▶ **Social signals:** Shares, mentions, and engagement on social media — though debated as a direct ranking factor, social signals correlate with brand authority and backlink acquisition.
- ▶ **Local SEO (Google Business Profile):** For businesses with physical locations or local delivery (Chaldal for Dhaka grocery), optimizing Google Business Profile ensures appearance in local map pack results — "grocery delivery near me."

TECHNICAL SEO FACTORS

Technical SEO ensures search engines can crawl and index your site efficiently: XML sitemap submission, robots.txt configuration, canonical tags (preventing duplicate content penalties), HTTPS implementation, fixing broken links (404 errors), and ensuring no orphaned pages. These are infrastructure-level requirements that a developer implements.

1B. SEARCH ENGINE MARKETING (SEM) / PAY-PER-CLICK (PPC)

SEM is paid advertising on search engines. Merchants bid on keywords; their ads appear at the top of search results (labeled "Sponsored" or "Ad"). Unlike SEO, results are immediate — ads appear as soon as the campaign is activated and payment is set up.

PPC MODEL — PAY PER CLICK

In the **Pay-Per-Click** model, the advertiser pays the search engine (Google, Bing) only when a user actually clicks on the ad — not for merely displaying it. The cost per click is determined by a real-time auction system where advertisers bid on keywords.

$$CPC = \text{Ad Spend} \div \text{Number of Clicks}$$

// Cost Per Click — how much each click to the website costs

$\text{Ad Rank} = \text{Bid Amount} \times \text{Quality Score}$

// Your ad position depends on both bid AND relevance/quality

$\text{Quality Score} = f(\text{CTR}, \text{Ad Relevance}, \text{Landing Page Quality})$

// Google grades your ad's relevance from 1-10; higher score = lower CPC

Key SEM concepts:

- ▶ **Keyword match types:** Broad match (ad shows for related searches), Phrase match (ad shows for searches containing the phrase), Exact match (ad shows only for that exact query). Broad match generates more volume but less relevant traffic; exact match is targeted but limited reach.
- ▶ **Negative keywords:** Terms for which you explicitly do NOT want your ad to appear. A seller of premium laptops adds "cheap" and "free" as negative keywords to avoid wasting budget on price-insensitive queries.
- ▶ **Ad extensions:** Additional information appended to ads — sitelink extensions (links to specific pages), call extensions (phone number), location extensions (store address), price extensions (product prices). Increase ad real estate and CTR.
- ▶ **Remarketing / Retargeting:** Ads targeted specifically at users who previously visited your website but did not purchase. These users have shown intent — they are 70% more likely to convert than cold traffic. A user who viewed a laptop on Daraz sees ads for that laptop across other websites and social media for the next 30 days.
- ▶ **Google Shopping Ads (PLAs — Product Listing Ads):** Visual ads showing product image, title, price, and store name directly in search results. Critical for e-commerce — users see the product and price before clicking.

1C. EMAIL MARKETING

Email remains the highest-ROI digital marketing channel — returning approximately \$36–\$42 for every \$1 spent (DMA data). It is a direct, owned channel — unlike social media where algorithm changes can destroy your reach overnight.

- ▶ **Transactional emails:** Order confirmations, shipping notifications, password resets — automatically triggered by user actions. Open rates are 3–5x higher than promotional emails because users expect and want them.
- ▶ **Promotional emails:** Sale announcements, new product launches, seasonal campaigns. Subject line and preview text determine open rate. Personalization ("Hi Rahim, based on your last order...") increases engagement 6x.
- ▶ **Abandoned cart emails:** Sent when a user adds items to cart but does not checkout. Typically 3-email sequence: reminder (1 hour later), social proof/reviews (24 hours), discount offer (72 hours). Recovers 5–15% of abandoned carts. Critical revenue recovery for e-commerce.

- ▶ **Win-back campaigns:** Emails targeting customers who have not purchased in 90+ days. Offer an incentive to re-engage.
- ▶ **Email segmentation:** Dividing the email list into groups based on behavior (high-value customers, first-time buyers, product category interests) and sending tailored messages. Segmented campaigns achieve 14% higher open rates and 100% higher click rates than unsegmented.

1D. SOCIAL MEDIA MARKETING

- ▶ **Organic social:** Free content posted to brand pages/profiles — product showcases, behind-the-scenes, user-generated content sharing, engagement with comments. Organic reach has declined dramatically on Facebook (now ~2–5% of followers see posts without paid promotion).
- ▶ **Paid social:** Facebook/Instagram ads — most sophisticated targeting in digital advertising. Target by demographics (age, gender, location), interests (technology, cooking), behavior (recent shoppers, frequent travelers), lookalike audiences (people similar to your existing customers). Facebook pixel tracks website actions for attribution and retargeting.
- ▶ **Influencer marketing:** Partnering with individuals who have engaged follower audiences. Categories: Nano influencers (1K–10K followers, very high engagement), Micro (10K–100K, niche authority), Macro (100K–1M), Mega/Celebrity (1M+). Bangladesh has a large micro-influencer economy particularly on Facebook and Instagram for fashion, beauty, and food.
- ▶ **Community management:** Responding to comments, messages, and reviews. Speed of response is a key trust signal — customers expect responses within 1–2 hours on social platforms.

1E. CONTENT MARKETING

Creating and distributing valuable, relevant content to attract and engage a target audience, rather than directly promoting products. Content marketing builds trust, authority, and long-term organic traffic.

- ▶ **Blog posts and buying guides:** "Best Laptops Under 50,000 Taka in 2024" — ranks for informational searches, attracts buyers in research phase, and includes product links generating sales. Chaldal's recipe content drives grocery purchases.
- ▶ **Video content:** Product demonstrations, unboxing, tutorials. YouTube is the second largest search engine — video content ranks for product queries. TikTok video content drives impulse purchases.
- ▶ **User-generated content (UGC):** Customer photos, reviews, unboxing videos. More trusted than brand-produced content. Daraz encourages buyers to post product photos in reviews — this UGC improves conversion for future buyers.

2. Key E-Commerce Metrics and Formulas

These metrics are the language of e-commerce performance measurement. Every formula must be memorized and understood conceptually.

CONVERSION RATE (CR)

$$CR = (\text{Purchases} \div \text{Visits}) \times 100$$

Percentage of website visitors who complete a purchase. Industry average: 1–4%. A 1% CR means 1 in every 100 visitors buys. Improving CR is the highest-leverage activity in e-commerce — doubling CR from 1% to 2% doubles revenue with the same traffic.

COST PER CLICK (CPC)

$$CPC = \text{Ad Spend} \div \text{Clicks}$$

Average cost of one click in a paid advertising campaign. If you spend 10,000 BDT on Google Ads and receive 500 clicks, $CPC = 20$ BDT/click. Lower CPC + higher CR = more profitable campaigns.

CUSTOMER ACQUISITION COST (CAC)

$$CAC = \text{Total Marketing Spend} \div \text{New Customers Acquired}$$

Total cost to acquire one new paying customer. If marketing spend is 100,000 BDT and 200 new customers were acquired, $CAC = 500$ BDT. Must be compared to CLV — if $CAC > CLV$, the business is unprofitable.

CUSTOMER LIFETIME VALUE (CLV)

$$CLV = AOV \times \text{Purchase Frequency} \times \text{Customer Lifespan}$$

Total revenue expected from one customer over their entire relationship with the business. If $AOV = 2,000$ BDT, frequency = 6 orders/year, lifespan = 3 years: $CLV = 2,000 \times 6 \times 3 = 36,000$ BDT. CLV:CAC ratio should be $\geq 3:1$ for a healthy e-commerce business.

AVERAGE ORDER VALUE (AOV)

$$AOV = \text{Total Revenue} \div \text{Number of Orders}$$

Average amount spent per transaction. If total revenue = 500,000 BDT from 250 orders, $AOV = 2,000$ BDT. Increasing AOV (through upselling, cross-selling, bundle offers, free shipping thresholds) directly increases CLV without increasing CAC.

RETURN ON AD SPEND (ROAS)

$$ROAS = \text{Revenue from Ads} \div \text{Ad Spend}$$

Revenue generated per taka spent on advertising. ROAS of 4 means every 1 BDT spent on ads returns 4 BDT in revenue. Minimum viable ROAS depends on product margin — a 25% margin product needs $ROAS > 4$ just to break even on ad spend.

BOUNCE RATE

$$BR = (\text{Single-Page Visits} \div \text{Total Visits}) \times 100$$

Percentage of visitors who leave after viewing only one page. High bounce rate ($>70\%$) indicates landing page irrelevance, slow page load, or poor mobile experience. E-commerce product

CART ABANDONMENT RATE

$$CAR = (1 - \text{Orders} \div \text{Cart Additions}) \times 100$$

Percentage of shoppers who add items to cart but do not complete the purchase. Industry average: $\sim 70\%$. Caused by: unexpected shipping costs, forced account creation, complex checkout, payment failures, price

pages should aim for <40% bounce rate.

comparison. Reducing CAR by 5% can increase revenue by 25%+.

EXAM TIP — NUMERIC CALCULATION QUESTIONS

You will almost certainly be asked to calculate at least one of these metrics in the exam. Practice: "An e-commerce business spends 50,000 BDT/month on marketing and acquires 100 new customers. Customers order 4 times/year, average order value is 1,500 BDT, and the average customer stays for 2 years. Calculate CAC, CLV, and the CLV:CAC ratio." Answer: CAC = 500 BDT; CLV = $1,500 \times 4 \times 2 = 12,000$ BDT; CLV:CAC = 24:1 (excellent).

3. The AIDA Funnel Model

AIDA FRAMEWORK

The **AIDA model** (Awareness → Interest → Desire → Action) describes the cognitive and emotional stages a consumer passes through from first encountering a brand to making a purchase. It is a marketing model that explains the steps a customer goes through before buying a product. Each stage requires different marketing tactics and content types.

A — Awareness

SEO, Social Ads, Display Ads, Influencers, PR, Viral Content

I — Interest

Blog content, Email newsletter, Retargeting, Product videos, Comparison guides

D — Desire

Reviews/testimonials, Limited-time offers, Personalization, Live chat, Social proof

A — Action

Optimized checkout, Multiple payment options, Trust badges, Clear CTA

Stage	Consumer State	Goal of Marketer	Digital Channels & Tactics	Key Metrics
Awareness	Consumer does not yet know the brand or product exists	Create brand recognition and reach the target audience	SEO (blog/informational content), social media ads (broad targeting), display advertising, influencer partnerships, PR/news coverage, viral marketing	Impressions, Reach, Brand search volume, New website visitors

Stage	Consumer State	Goal of Marketer	Digital Channels & Tactics	Key Metrics
Interest	Consumer is aware but evaluating options, learning more	Educate and engage; establish authority and relevance	Email newsletter subscriptions, retargeting ads to visitors, in-depth product content (how-to videos, buying guides, comparison articles), webinars, free trials, demos	Time on site, Pages per session, Email sign-up rate, Return visitor rate, Content engagement
Desire	Consumer wants the product but has not yet decided to buy from you	Convert want into conviction and preference for your brand	Customer testimonials and reviews, case studies, social proof (number of buyers, star ratings), scarcity tactics ("Only 3 left in stock"), urgency ("Sale ends tonight"), abandoned cart emails, live chat support, personalized product recommendations	Add-to-cart rate, Wishlist additions, Product page dwell time, Review engagement
Action	Consumer is ready to purchase	Remove all friction from the purchase process and close the sale	Streamlined checkout (minimal steps), guest checkout option, multiple payment methods (bKash, card, COD), clear shipping info and delivery times, trust badges (SSL certificate, payment logos, return policy), prominent call-to-action (CTA) buttons	Conversion Rate, Cart Abandonment Rate, Revenue, Average Order Value, Cost Per Acquisition

EXTENDED FUNNEL — BEYOND AIDA

Modern digital marketing extends the funnel beyond the initial purchase to retention and advocacy:

- ▶ **Retention (Loyalty):** Post-purchase email nurturing, loyalty points programs (Daraz Coins), personalized recommendations based on purchase history, subscription programs (free delivery for loyal customers). Retaining an existing customer costs 5–7x less than acquiring a new one.
- ▶ **Advocacy (Referral):** Turning satisfied customers into brand ambassadors who refer friends. Referral programs (earn credit for referring), review requests, community building. Word-of-mouth is the highest-trust acquisition channel.

4. A/B Testing for Conversion Rate Optimization

A/B TESTING (SPLIT TESTING)

A/B testing is a controlled experiment in which two versions of a web page, email, or ad (Version A and Version B) are shown to different randomly assigned segments of users simultaneously, and the version

that produces the better outcome (higher conversion rate, more clicks, more revenue) is selected. It is the scientific method applied to digital marketing optimization.

A/B TESTING PROCESS

1 Identify Hypothesis

Based on data (heatmaps, session recordings, analytics), identify a specific element to test with a clear hypothesis. Example: "Changing the CTA button color from gray to green will increase add-to-cart rate because green is associated with positive action and will stand out more."

2 Create Variations

Version A = the existing "control" page. Version B = the modified "variant" — only one element changed at a time (button color, headline text, image, price display format, CTA copy). Testing multiple changes simultaneously makes it impossible to know which change caused the result — this is called multivariate testing and requires much larger sample sizes.

3 Split Traffic Randomly

Incoming users are randomly assigned to either Version A or B — typically 50/50. The random assignment is critical to eliminate bias. Same users see the same version consistently (cookie-based assignment) to avoid confusion.

4 Collect Data Until Statistical Significance

The test must run long enough to accumulate sufficient conversions for results to be statistically reliable — not due to random chance. A minimum of 95% statistical confidence is the industry standard. Stopping tests early ("peeking") when one version looks better is a common mistake — premature conclusions are often wrong.

5 Implement Winner and Iterate

Deploy the winning version to all users. Document the result and the insight learned. Run the next test. CRO is a continuous process — Amazon runs thousands of simultaneous A/B tests. Even small improvements (0.5% CR increase) compound significantly at scale.

COMMON A/B TESTING ELEMENTS IN E-COMMERCE

- ▶ **Landing page headline:** "Free Delivery on All Orders" vs. "Order Now — Delivered Tomorrow"
- ▶ **CTA button text and color:** "Buy Now" vs. "Add to Cart" vs. "Get It Today"; Blue vs. Orange button
- ▶ **Product image:** White background product shot vs. lifestyle/contextual image
- ▶ **Price display:** "2,000 BDT" vs. "Only 1,999 BDT" vs. "Save 500 BDT — Was 2,500"
- ▶ **Social proof placement:** Star rating near the top of the page vs. below the fold

- ▶ **Checkout flow:** Single-page checkout vs. multi-step checkout wizard
-

- ▶ **Free shipping threshold:** "Free shipping over 500 BDT" vs. "Free shipping over 1,000 BDT"

EXAM TIP

A/B testing is a high-value exam topic because it combines the concepts of digital marketing optimization, measurement, and decision-making. A likely exam question: "What is A/B testing? Explain with steps how an e-commerce company can use it to improve its conversion rate. Give two examples of elements that could be tested." Know the 5-step process and at least 3 testable elements.

E-commerce logistics is what turns a digital transaction into a physical delivery. It is widely recognized as the most operationally complex and capital-intensive aspect of e-commerce. For every dollar customers spend online, a significant fraction covers the cost of getting the product from a warehouse to their doorstep. Efficient logistics is a core competitive differentiator — Amazon Prime's 2-day (now same-day in many US cities) delivery is a strategic moat that competitors struggle to match.

1. Last-Mile Delivery

LAST-MILE DELIVERY

Last-mile delivery refers to the final step in the delivery chain — moving a parcel from a local distribution hub or delivery station to the end customer's address. Despite being the shortest segment of the journey geographically, last-mile delivery accounts for **53% of total shipping costs** and is the most complex, labor-intensive, and customer-visible part of e-commerce fulfillment.

WHY LAST-MILE IS SO EXPENSIVE AND CHALLENGING

- ▶ **Disaggregation:** A truck driving from Chattogram to Dhaka carries hundreds of parcels efficiently. But those parcels must then be split into individual deliveries to hundreds of different addresses — each requiring a separate vehicle trip. This disaggregation is extremely inefficient.
- ▶ **Failed delivery attempts:** Customers are not always home. A failed delivery requires a re-attempt (additional cost) or a return (significant cost). Failed deliveries cost couriers \$17 on average globally per failed attempt.
- ▶ **Address quality in Bangladesh:** Bangladesh lacks standardized postal addressing. Many addresses are descriptive ("near the blue mosque, behind the school") rather than structured (street number, postcode). GPS accuracy in rural areas is poor. Delivery agents spend significant time locating addresses.
- ▶ **Density economics:** Urban areas (Dhaka, Chattogram) have high delivery density — many deliveries per km² — making last-mile efficient. Rural areas are very low-density, making each delivery disproportionately expensive. This creates a rural delivery affordability challenge.
- ▶ **Traffic congestion:** Dhaka has among the world's worst traffic congestion. A delivery planned for 4 hours can take 8 hours in traffic. Congestion increases cost, reduces delivery capacity per driver per day, and creates unpredictability in delivery time windows.
- ▶ **Customer expectation escalation:** Amazon Prime trained customers to expect next-day or same-day delivery. Meeting these expectations requires expensive local warehousing near population centers (dark

stores, micro-fulfillment centers).

ROUTE OPTIMIZATION (CONCEPTUAL)

Route optimization uses algorithms (Travelling Salesman Problem heuristics, Vehicle Routing Problem solutions) to determine the most efficient sequence of stops for a delivery vehicle to minimize total distance, time, and fuel cost. Inputs: delivery addresses, vehicle capacity, time windows, traffic data. Modern systems (Google Maps Platform, HERE Technologies) solve these problems in real time. Pathao's logistics division uses algorithmic routing for its Dhaka deliveries.

LAST-MILE INNOVATIONS

- ▶ **Parcel lockers (PUDO — Pick Up Drop Off points):** Secure lockers at convenient locations (petrol stations, supermarkets) where customers collect parcels at their own convenience. Eliminates the "not at home" problem. Amazon Lockers are the dominant example. Bangladesh: some courier companies piloting collection points at convenience stores.
- ▶ **Crowd-sourced delivery (gig delivery):** Platforms like Lalamove, Pathao Delivery use gig economy workers (motorcycle couriers) for on-demand same-day delivery. Flexible, scalable, but quality control challenges.
- ▶ **Drone delivery (experimental):** Amazon Prime Air, Wing (Alphabet) testing drone delivery in suburban areas for small parcels. Not yet commercially viable at scale — regulatory, safety, and range challenges.
- ▶ **Autonomous delivery vehicles:** Self-driving ground robots for sub-5km delivery in controlled environments (university campuses, business parks). Startups: Starship Technologies, Nuro.

2. Fulfillment Models

How products are stored, packed, and delivered to customers after an order is placed.

2A. IN-HOUSE WAREHOUSING (FIRST-PARTY FULFILLMENT)

The merchant owns or leases warehouse space, maintains their own inventory, employs warehouse staff, and directly manages packing and dispatching orders. Full control over the fulfillment process and customer experience. Amazon and Daraz's merchant warehouses are examples.

- ▶ **Advantages:** Full control over packaging, branding, quality; fastest response to issues; no dependency on third parties; can offer same-day dispatch.
- ▶ **Disadvantages:** High fixed costs (lease, staff, equipment); capital tied up in inventory; requires expertise in warehouse management; limited scalability without significant investment.
- ▶ **Best for:** Merchants with high, predictable volume; premium brands requiring branded packaging; businesses with perishable inventory (Chaldal's fresh grocery).

2B. DROPSHIPING

DROPSHIPPING MODEL

In **dropshipping**, the merchant does not hold any inventory. When a customer places an order, the merchant forwards the order to a third-party supplier (manufacturer or wholesaler), who ships the product directly to the customer. The merchant never sees or touches the physical product.

Feature	Dropshipping	In-House Warehousing
Upfront capital required	Very low — no inventory purchase required	High — must purchase and store inventory
Margin	Lower — supplier takes their margin; merchant earns difference	Higher — merchant buys at wholesale and earns full retail margin
Product control	None — supplier decides product quality and packaging	Full control over product quality and presentation
Shipping time	Potentially longer — especially if supplier is overseas (China)	Fast — merchant can offer same-day dispatch
Scalability	Highly scalable — adding products costs nothing	Scaling requires more warehouse space and capital
Risk	Stock availability not guaranteed; supplier errors damage your reputation	Overstocking risk (capital tied up); understocking risk (lost sales)
Best for	New e-commerce businesses testing product-market fit; niche product merchants; catalog-wide variety sellers	Established merchants, branded products, fast-moving categories

2C. 3PL — THIRD-PARTY LOGISTICS

The merchant outsources some or all logistics functions to a specialized third-party logistics provider (3PL): warehousing, inventory management, order picking and packing, shipping, and returns processing. The merchant ships bulk inventory to the 3PL's warehouses; the 3PL handles individual order fulfillment.

Examples: *Amazon FBA (Fulfillment by Amazon)* — the world's largest 3PL. Sellers ship inventory to Amazon warehouses; Amazon stores, picks, packs, ships, and handles returns. Products become Prime-eligible. In Bangladesh: *Shajgoj's warehouse + fulfillment services for beauty brands*, *Pathao Fulfillment*.

- **Advantages:** Access to professional logistics infrastructure without capital investment; economies of scale; 3PL expertise in warehouse operations; allows merchant to focus on core business (product development, marketing).
- **Disadvantages:** Loss of direct control over fulfillment quality; 3PL errors affect your brand reputation; storage fees accumulate for slow-moving inventory; minimum volume requirements with most 3PLs.

2D. 4PL — FOURTH-PARTY LOGISTICS

A 4PL is a logistics integrator that manages the entire supply chain on behalf of the client — orchestrating multiple 3PLs, carriers, and technology platforms into a seamless end-to-end solution. The 4PL is essentially a supply chain management outsourcing provider. Used by large enterprises with complex, global supply chains.

3. Inventory Management and the EOQ Formula

INVENTORY MANAGEMENT

Inventory management is the process of ordering, storing, tracking, and controlling a merchant's stock of goods. The central challenge: hold *enough* inventory to meet demand without incurring excess holding costs, and avoid *too little* inventory that results in stockouts and lost sales. This balance between over-stocking and under-stocking is the fundamental inventory optimization problem.

EOQ — ECONOMIC ORDER QUANTITY FORMULA

The **Economic Order Quantity (EOQ)** model determines the optimal quantity to order at a time that minimizes the total inventory cost (ordering cost + holding cost). It balances two opposing cost forces: ordering more at once reduces the cost per order but increases holding/storage costs; ordering less frequently reduces holding costs but increases ordering frequency and cost.

$$EOQ = \sqrt{(2DS / H)}$$

Where:

D = Annual Demand (units per year)

S = Ordering Cost per order (cost to place one purchase order)

H = Holding Cost per unit per year (storage, insurance, obsolescence)

EOQ WORKED EXAMPLE

An e-commerce electronics seller sells 1,200 units/year of a specific power bank. Each purchase order costs 500 BDT to process (staff time, admin). Holding one unit in the warehouse costs 50 BDT/year (rent, electricity, insurance).

$$D = 1,200 \text{ units/year} \quad S = 500 \text{ BDT/order} \quad H = 50 \text{ BDT/unit/year}$$

$$EOQ = \sqrt{(2 \times 1,200 \times 500 / 50)} \quad EOQ = \sqrt{(1,200,000 / 50)} \quad EOQ = \sqrt{24,000} \quad EOQ \approx 155 \text{ units per order}$$

This means the optimal order size is approximately 155 units. The seller should order 155 units at a time, placing approximately $1,200/155 \approx 7\text{--}8$ orders per year. Ordering less than 155 units means ordering too frequently (high ordering cost); ordering more means holding too much stock (high holding cost).

EOQ ASSUMPTIONS AND LIMITATIONS

- ▶ **Assumes constant, known demand:** In reality, e-commerce demand fluctuates seasonally (Eid sales spike), rendering a fixed EOQ suboptimal. EOQ must be recalculated periodically.
- ▶ **Assumes constant ordering and holding costs:** Supplier price breaks (bulk discounts) change the economics significantly. A separate model (Quantity Discount Model) handles this.
- ▶ **Ignores stockouts:** The basic EOQ model assumes demand is always met — it does not account for the cost of running out of stock (lost sales, customer dissatisfaction).
- ▶ **Ignores lead time:** In practice, orders are placed before stock runs out, incorporating a reorder point based on lead time and safety stock. $\text{Reorder Point} = (\text{Daily Demand} \times \text{Lead Time in days}) + \text{Safety Stock}$.

OTHER INVENTORY MANAGEMENT APPROACHES

- ▶ **Just-In-Time (JIT):** Inventory arrives exactly when needed for production or sale, minimizing holding costs. Requires reliable, fast suppliers. Vulnerable to supply chain disruptions (COVID-19 showed this risk dramatically).
- ▶ **ABC Analysis:** Classifying inventory into three categories: A (high value, 20% of SKUs but 80% of revenue — manage most tightly), B (medium value), C (low value, 50% of SKUs but only 5% of revenue — minimal management). Focus management attention where it matters most.
- ▶ **Safety Stock:** Buffer inventory held above the minimum needed, as protection against demand spikes or supplier delays. $\text{Safety Stock} = Z \times \sigma \times \sqrt{\text{Lead Time}}$, where Z is the desired service level z-score and σ is the standard deviation of daily demand.
- ▶ **Perpetual vs. Periodic Inventory:** Perpetual inventory systems update stock counts in real-time as sales occur (via POS integration and barcode scanning). Periodic systems count inventory at scheduled intervals. E-commerce requires perpetual systems to prevent overselling.

4. Returns Management (Reverse Logistics)

REVERSE LOGISTICS

Reverse logistics refers to all processes involved in moving goods from the end customer back to the seller or manufacturer — for returns, refurbishment, recycling, or disposal. It is the "reverse" of the normal supply chain flow. E-commerce return rates are dramatically higher than physical retail: apparel returns average 20–30% online vs. 8–10% in-store; electronics returns average 10–15%.

WHY RETURNS ARE EXPENSIVE

- ▶ **Reverse transportation cost:** Return shipping costs must be absorbed (by the merchant or shared). Free returns are an expectation in competitive markets, but significantly increase cost.

- ▶ **Processing labor:** Every returned item must be received, inspected, categorized (resaleable, damaged, needs refurbishment), restocked or disposed of. Labor-intensive.
- ▶ **Restocking or disposal:** Returned items may be damaged, opened, or date-expired — not resaleable at full price. May need to be sold as "refurbished," sold in a secondary market, or disposed of (environmental cost).
- ▶ **Inventory accuracy disruption:** Returns arrive at unpredictable times and must be processed back into inventory. This temporarily distorts inventory counts.
- ▶ **Fraud:** Return fraud (returning worn items, returning counterfeit items, "wardrobing" — buying and returning after use) is estimated at 5–8% of all returns in developed markets.

RETURNS MANAGEMENT BEST PRACTICES

- ▶ **Clear, easy returns policy:** Amazon's no-questions-asked 30-day return policy is a conversion-rate driver — customers buy more confidently knowing returns are easy. Daraz's 7-day return policy is standard in Bangladesh.
- ▶ **Self-service returns portal:** Allow customers to initiate returns online, print labels, and track refund status without calling customer service. Reduces support costs.
- ▶ **Return reason analysis:** Systematically collecting return reasons enables product improvement (high return rate for "product not as described" indicates misleading listings) and operational improvement (high "wrong item sent" rate indicates picking errors).
- ▶ **Refurbishment and secondary channels:** Returned-but-functional items can be sold as "open box" or "certified refurbished" at a discount, recovering partial value instead of full disposal.

5. Role of RFID and IoT in Supply Chain Visibility

RFID — RADIO FREQUENCY IDENTIFICATION

RFID

RFID uses electromagnetic fields to automatically identify and track tags attached to objects. An RFID tag (a microchip with an antenna) can be read by an RFID reader from several meters away, without requiring line-of-sight. Unlike barcodes, RFID tags can be read in bulk (100+ tags simultaneously), are read/write capable, and can store more information.

- ▶ **RFID in warehouse management:** Products tagged with RFID chips can be automatically counted as they move through warehouse receiving docks, without manual scanning. Real-time inventory accuracy improves from ~65% (manual counting) to ~99.5% (RFID-enabled).

- ▶ **RFID in retail:** Walmart mandated RFID tagging on all pallet-level shipments from suppliers in 2003 — one of the most influential supply chain technology decisions. Item-level RFID in apparel retail (Zara, H&M) enables accurate real-time stock counts, reducing stockouts and overstock.
- ▶ **RFID for anti-theft:** Electronic Article Surveillance (EAS) RFID tags on retail merchandise trigger alarms if not deactivated at checkout. Widespread in fashion and electronics retail.
- ▶ **Passive vs. active RFID:** Passive tags have no internal power source — they draw energy from the reader's electromagnetic field. Range: up to 10 meters. Cheap (fractions of a dollar per tag). Active tags have their own battery — range up to 100m, higher cost. Used for tracking expensive assets (shipping containers, vehicles, expensive equipment).

IOT — INTERNET OF THINGS IN SUPPLY CHAIN

IOT DEFINITION

The **Internet of Things (IoT)** refers to the network of physical objects embedded with sensors, software, and connectivity that enables them to collect and exchange data over the internet. In supply chain, IoT provides real-time visibility into the location, condition, and status of goods throughout their journey.

- ▶ **GPS tracking on delivery vehicles:** Real-time location of every delivery truck, motorcycle, or van. Enables accurate ETAs for customers, route optimization, driver performance monitoring, and unauthorized use detection. Pathao Courier and Paperfly use GPS tracking in Bangladesh.
- ▶ **Temperature and humidity sensors (Cold chain monitoring):** Critical for perishable goods — food, pharmaceuticals, vaccines. IoT sensors continuously monitor and transmit temperature data inside refrigerated trucks and cold storage facilities. Threshold breaches trigger instant alerts. Chaldal's cold chain for fresh produce relies on temperature IoT monitoring.
- ▶ **Smart shelves:** Retail shelves equipped with weight sensors that detect when product levels fall below a threshold, automatically triggering a restocking alert. Eliminates the need for manual shelf audits.
- ▶ **Warehouse automation with IoT:** Amazon's Kiva robots (now Amazon Robotics) navigate warehouse floors using IoT positioning systems, bringing product shelves to human pickers rather than having workers walk miles of aisles. Amazon has 750,000+ robots in its warehouses globally.
- ▶ **Port and container tracking:** IoT sensors on shipping containers track location, temperature, humidity, shock/vibration, and tampering throughout oceanic transit. Critical for Bangladesh's garment export supply chains where container conditions affect product quality claims.

6. Cross-Border E-Commerce

Cross-border e-commerce refers to online sales transactions between buyers and sellers in different countries. It represents the fastest-growing segment of global e-commerce, but also the most operationally complex.

KEY CHALLENGES

- ▶ **Customs and import duties:** Every country has its own tariff schedule. Products entering Bangladesh are subject to NBR (National Board of Revenue) import duties — the rate varies by product category. Merchants must correctly classify products using HS (Harmonized System) codes. Incorrect classification causes customs holds and additional charges. DDP (Delivered Duty Paid) vs. DDU (Delivered Duty Unpaid) terms define who pays — DDP provides better customer experience but requires complex landed cost calculation upfront.
- ▶ **International shipping carriers and costs:** DHL Express, FedEx, Aramex, and EMS (Bangladesh Post) handle international parcels. Costs are significantly higher than domestic — a 1kg parcel from Dhaka to Germany via DHL costs approximately \$25–\$40. Dimensional weight pricing (volumetric weight) applies: the carrier charges the greater of actual weight and volumetric weight ($L \times W \times H \div 5000$).
- ▶ **Export documentation:** Commercial invoice, packing list, certificate of origin, export permit (for controlled goods), fumigation certificates (for wood packaging). Missing documents cause customs delays. For Bangladesh garment exports: also requires factory compliance certificates (BGMEA certificates), buyer-specific documentation.
- ▶ **Currency and payment:** International buyers may not have access to the merchant's local payment methods. Payment gateways like Stripe, PayPal, and 2Checkout support multi-currency card payments. Dynamic currency conversion shows prices in the buyer's local currency.
- ▶ **Language and localization:** Product listings, customer support, and legal terms must be translated. Regulatory labeling requirements differ — EU requires CE marking, food products require ingredient labels in local language.
- ▶ **Returns across borders:** International returns are extremely expensive — often costing more than the product itself. Most cross-border merchants offer in-country return solutions (partner warehouses in destination countries) or simple "keep the item" refund policies for low-value items.

7. Case Study: Alibaba's Cainiao Network

CASE STUDY — CAINIAO SMART LOGISTICS NETWORK

Alibaba does not own its own logistics network the way Amazon does. Instead, it built **Cainiao** (菜鸟) as a technology and data platform that orchestrates hundreds of third-party logistics companies — carriers, warehouse operators, customs brokers, and last-mile couriers — into a unified, intelligently coordinated network. Cainiao is a 4PL operating at the world's largest scale.

HOW CAINIAO WORKS

- ▶ **Data integration layer:** Cainiao's platform integrates with every logistics partner's system via APIs, giving Alibaba (and merchants) a unified view of every parcel's location and status across all carriers.
- ▶ **Intelligent order routing:** When a merchant ships an order, Cainiao's algorithm analyzes destination, package weight/dimensions, required delivery speed, carrier capacity, and cost — then automatically routes the order to the optimal carrier. This "smart routing" reduces cost and improves delivery time simultaneously.
- ▶ **Warehouse network:** Cainiao operates bonded warehouses (in free trade zones) in China and bonded logistics centers in destination countries. International merchants can ship in bulk to Cainiao's overseas bonded warehouses — goods clear customs once for many orders, dramatically reducing per-order customs cost and time.
- ▶ **Cainiao Post (pickup network):** 170,000+ community collection points across China where rural/apartment residents can collect parcels. Addresses the "last-mile access" problem at enormous scale.
- ▶ **Singles' Day scale:** During 11.11 2023, Cainiao processed 1 billion+ packages. This required predictive demand modeling (warehousing stock near predicted high-demand areas before the sale), pre-sorting of orders during the pre-order phase, and massive surge capacity activation across hundreds of logistics partners simultaneously.
- ▶ **Cross-border:** Cainiao eHub — dedicated international logistics hubs in Liège (Belgium), Kuala Lumpur (Malaysia), Dubai (UAE), and other cities, providing fast, direct air freight connections for cross-border commerce, bypassing slow postal systems.

STRATEGIC LESSON FROM CAINIAO

Cainiao demonstrates that in e-commerce logistics, technology and data coordination can be more powerful than owning physical assets. By creating a platform that makes hundreds of logistics companies more efficient and interoperable, Alibaba achieves Amazon-level logistics capability at a fraction of the capital investment. This "asset-light, data-heavy" model is increasingly influential in supply chain management globally.

8. Summary: Key Concepts for Exam Review

Concept	Core Definition	Key Formula / Example
Last-Mile Delivery	Final leg from local hub to customer; most expensive part of delivery	53% of shipping cost; solutions: PUDO, route optimization, crowd-sourced delivery

Concept	Core Definition	Key Formula / Example
In-House Fulfillment	Merchant owns and operates own warehouse and dispatch	Full control but high fixed cost; Chaldal grocery model
Dropshipping	Merchant sells without holding inventory; supplier ships directly	Low capital; lower margin; no control over shipping speed/quality
3PL	Third-party specialist handles warehousing and fulfillment	Amazon FBA; Pathao Fulfillment; economies of scale
4PL	Manages and coordinates multiple 3PLs and the entire supply chain	Cainiao; supply chain as a service
EOQ	Optimal order quantity minimizing total inventory cost	$EOQ = \sqrt{(2DS/H)}$; D=demand, S=order cost, H=holding cost/unit
Reverse Logistics	Process of handling customer returns	20–30% return rate in apparel e-commerce; free returns drive conversion
RFID	Radio-frequency tags enabling automatic inventory tracking without line-of-sight	Walmart RFID mandate; 99.5% inventory accuracy; passive vs. active tags
IoT in supply chain	Connected sensors providing real-time visibility of goods in transit and storage	GPS tracking, temperature sensors, smart shelves, Amazon Robotics
Cross-border e-commerce	Online transactions between buyers and sellers in different countries	Challenges: customs/duties, shipping cost, documentation, returns, currency
Cainiao Network	Alibaba's logistics coordination platform orchestrating third-party carriers	1B+ packages on 11.11; bonded warehouses; smart routing algorithm

FINAL EXAM STRATEGY — CLASSES 04 TO 06

For Class 04: Be ready to draw and explain the three-tier architecture diagram; distinguish REST vs SOAP and HTTP vs HTTPS; explain CDN with an example; compare horizontal vs. vertical scaling. For Class 05: Memorize all metric formulas (CR, CAC, CLV, AOV, ROAS, CPC); know AIDA with examples of tactics at each stage; explain A/B testing process with steps. For Class 06: Calculate EOQ from given data; compare dropshipping vs. 3PL vs. in-house with pros and cons; explain last-mile challenges in Bangladesh context; describe what Cainiao is and why it is significant. The EOQ formula calculation is almost certainly in the exam.

